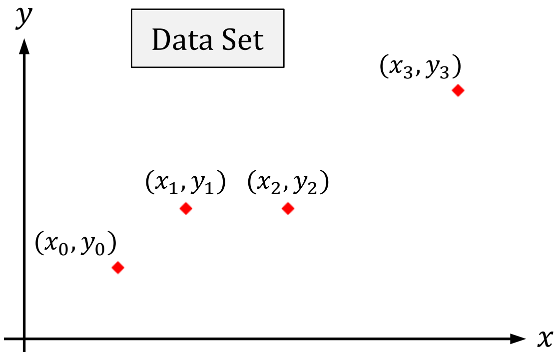
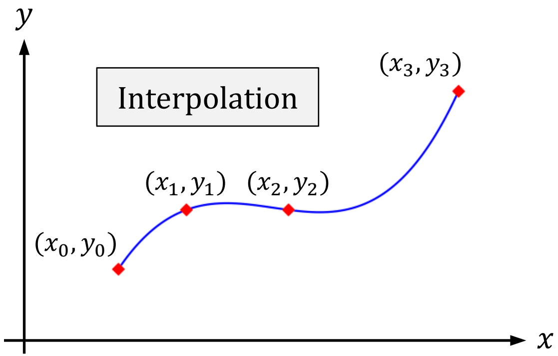
Concept & Outline



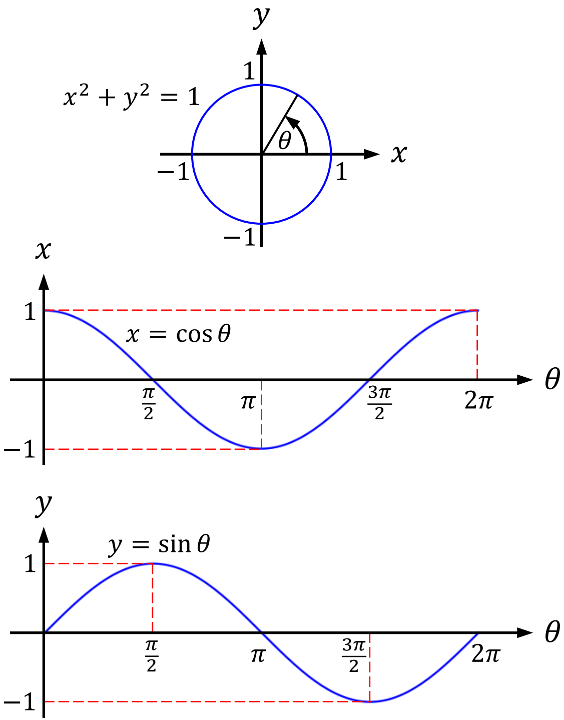


4 Curve Fitting

- **Newton Interpolating Polynomials**
  One function for all points.
  - Linear Interpolation
  - Quadratic Interpolation
  - General Form
- **Lagrange Interpolating Polynomials**
  One function for all points.
  - Linear Interpolation
  - Quadratic Interpolation
  - General Form
- **Spline Interpolation**
  Interpolation functions for an interval between two adjacent points.
  - Linear, Quadratic, and Cubic Spline
  - Modified Cubic Spline

Concept & Outline

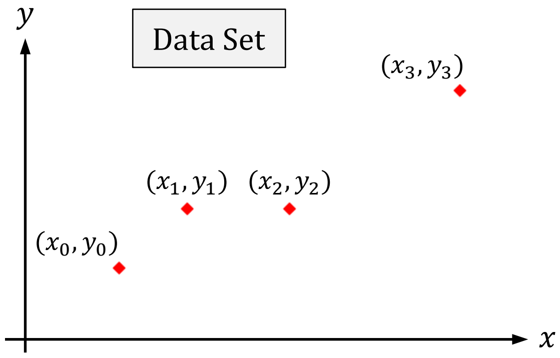
Parametric Curve

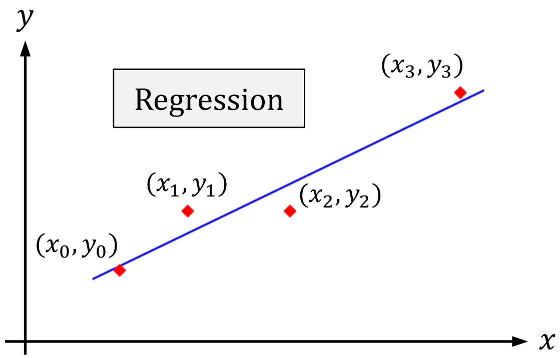


④ Curve Fitting

- **Newton Interpolating Polynomials**
  One function for all points.
  - Linear Interpolation
  - Quadratic Interpolation
  - General Form
- **Lagrange Interpolating Polynomials**
  One function for all points.
  - Linear Interpolation
  - Quadratic Interpolation
  - General Form
- **Spline Interpolation**
  Interpolation functions for an interval between two adjacent points.
  - Linear, Quadratic, and Cubic Spline
  - Modified Cubic Spline
- **Parametric Curves**
  Functions for coordinates of all points on a curve.

Concept & Outline





4 Curve Fitting

- Newton Interpolating Polynomials
  One function for all points.
  - Linear Interpolation
  - Quadratic Interpolation
  - General Form
- ✓ Lagrange Interpolating Polynomials
  One function for all points.
  - Linear Interpolation
  - Quadratic Interpolation
  - General Form
- Spline Interpolation
  Interpolation functions for an interval between two adjacent points.
  - Linear, Quadratic, and Cubic Spline
  - Modified Cubic Spline
- Parametric Curves
  Functions for coordinates of all points on a curve.

Interpolation & Extrapolation
Function(s) passing all points.

- ✓ Least Squares Regression
  One function for trend of all points.
  - Least Squares Regression by Polynomials
  - Multiple Least Squares Regression
  - Continuous Least Squares

Regression
Trend of all points or a function.